Nima Karimipour

Staff Software Engineer at Sonar · working on Gitar · Ph.D. in CS

NIMA.KARIMIPOUR@SONARSOURCE.COM · NIMAKARIMIPOUR.GITHUB.IO · GITHUB · LINKEDIN · GOOGLE SCHOLAR

SOFTWARE ENGINEERING · PROGRAMMING LANGUAGES · STATIC ANALYSIS · SECURITY AND RELIABILITY · AI AGENTS

Staff Software Engineer at Sonar, building agents on Gitar, the AI-native code review platform Sonar acquired in May 2026. Ph.D. in Computer Science from the University of California, Riverside, advised by Prof. Manu Sridharan. Research centers on programming languages and software engineering, with a focus on pluggable type systems, null safety, taint analysis, and automating code fixes with large language models (LLMs), building tools that boost the reliability and security of large-scale software systems.

EXPERIENCE

Staff Software Engineer
2026 – PRESENT

Sonar — Gitar

- Building agents for software engineering automation on Gitar, now part of Sonar's code verification platform.
- Gitar reviews code from the moment an agent starts writing it to the moment it lands, generating fixes and iterating until CI passes.

Research Scientist
2025 – 2026

Gitar.ai

- Built agents for software engineering automation.
- Gitar was acquired by Sonar in May 2026, expanding Sonar's code verification platform into AI code review.

Ph.D. Research Intern
JUN – SEP 2022

Uber Technologies, Inc. — Programming Systems Group (PSG)

- Enhanced NullAway, Uber's type-based checker designed to prevent NullPointerExceptions (NPEs) across Java and Android monorepos.
- Extended Annotator to work at monorepo scale, with support for build target boundaries so that inferred annotations do not trigger errors in downstream dependencies.
- Automated annotation inference over millions of lines of code, leading to the complete enrollment of Uber's Java monorepo into NullAway.
- Published the work with Uber at ESEC/FSE 2023: *Practical Inference of Nullability Types*.

EDUCATION

Ph.D. in Computer Science and Engineering
2020 – 2025

University of California, Riverside — Advisor: Prof. Manu Sridharan (RIPLE group)

B.Sc. in Computer Engineering
2014 – 2018

Sharif University of Technology, Tehran, Iran

PUBLICATIONS

LLM-Based Repair of Static Nullability Errors PDF
ISSTA 2026

Nima Karimipour, Pascal Joos, Michael Pradel, Martin Kellogg, and Manu Sridharan

35th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2026)

Practical Type-Based Taint Checking and Inference PDF ECOOP 2025

Nima Karimipour, Kanak Das, Behnaz Hassanshahi, and Manu Sridharan (in collaboration with Oracle Labs)

39th European Conference on Object-Oriented Programming (ECOOP 2025)

- Proposes a type-based taint checker that reduces false positives, improving handling of third-party libraries and complex language constructs.
- Introduces an automated technique to infer tainting type qualifiers, including support for polymorphic annotations and generic types.
- Outperforms whole-program taint analyzers (e.g. CodeQL), achieving comparable precision and recall while operating 27× faster.

A New Approach to Evaluating Nullability Inference Tools DOI FSE 2025

Nima Karimipour, Erfan Arvan, Martin Kellogg, and Manu Sridharan

33rd ACM International Conference on the Foundations of Software Engineering (FSE 2025)

- Identifies bias in prior evaluations of nullability inference tools, showing that "type reconstruction" experiments can inflate effectiveness.
- Proposes a new, unbiased definition of "best" inferred annotations, better suited for comparing inference tools.
- Conducts the first head-to-head comparison of three inference tools, revealing their unique strengths and areas for improvement.

Practical Inference of Nullability Types PDF · TALK ESEC/FSE 2023

Nima Karimipour, Justin Pham, Lazaro Clapp, and Manu Sridharan (in collaboration with Uber Technologies, Inc.)

31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE 2023)

- Introduces a tool to automatically infer nullability type qualifiers, easing NPE prevention in large Java codebases, using a black-box search strategy to maximize type-checked code.
- Achieves a 69.5% reduction in NullAway errors on average across Java projects.
- Deployed in production at Uber, enrolling 160+ modules and 1.3 million lines of code into NullAway.

OPEN SOURCE

Annotator *Java* 2023

Automated tool that integrates NullAway into existing codebases by inferring and applying nullability annotations at scale.

- Instantly generates patches with no remaining checker errors, applying suppressions where annotations alone cannot resolve them.
- Supports large-scale projects incrementally, target by target; scaled to fully integrate NullAway across Uber's entire monorepo.
- Recognized with a dedicated blog post by Gitar: *Automating Java Codebase Annotations for Null Safety*.

NullAway *Java* 2022

A fast, practical null pointer checker for Java, built as an Error Prone plugin. Core contributor with type inference extensions.

- Enhanced to support expressing method preconditions and postconditions.
- Added a serialization service to log errors and compute quick-fix suggestions for errors reported by the checker.

TaintTyper *Java* 2024

A practical type-based taint checker and inference tool for Java, exceeding recall of state-of-the-art whole-program taint analyzers.

- Modular and incremental checking for better scalability and performance.
- Automatically infers tainting type qualifiers, including polymorphic and generic type argument annotations.

AWARDS

Dean's Distinguished Fellowship Award 2020

Ph.D. program, University of California, Riverside